

Matrix Theory: Lecture 7

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Outline

- Functions on Matrices, $F : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$
- Definition using Jordan Form.
- Computation.



Functions on Matrices

We have studied polynomial expressions of matrices. If

$p(x) = x^k + a_{k-1}x^{k-1} + \dots + a_1x + a_0$ then for a (square) matrix X

$$p(X) = X^k + a_{k-1}X^{k-1} + \dots + a_1X + a_0I$$

Can we do the same for general functions? Ex. $\sin(x)$, $\cos(x)$, e^x ...

Taylor expansion:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

Each term $(x-a)^k$ is polynomial.

(Works but assumes that the series converges.

Want to avoid convergence issues.)



General Definition

Based on four rules (fulfilled for polynomials, and Taylor expansions):

- ① If $S^{-1}AS = J$ is the Jordan form then

$$f(A) = Sf(J)S^{-1}.$$

- ② If J is diagonal, $J = D = \text{diag}(\lambda_1, \dots, \lambda_n)$ then

$$f(D) = \text{diag}(f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)).$$

- ③ More generally, if

$$J = \begin{pmatrix} J_{k_1}(\lambda_1) & 0 & \cdots & 0 \\ 0 & J_{k_2}(\lambda_2) & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & J_{k_m}(\lambda_m) \end{pmatrix},$$

then

$$f(J) = \begin{pmatrix} f(J_{k_1}(\lambda_1)) & 0 & \cdots & 0 \\ 0 & f(J_{k_2}(\lambda_2)) & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & f(J_{k_m}(\lambda_m)) \end{pmatrix}.$$

- ④ $f(J_{k_i}(\lambda_i))$ defined by Taylor expansion.



Rule 4

$$J_k(\lambda) = \underbrace{\begin{pmatrix} \lambda & 0 & \dots & 0 & 0 \\ 0 & \lambda & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda & 0 \\ 0 & 0 & \dots & 0 & \lambda \end{pmatrix}}_{\lambda I} + \underbrace{\begin{pmatrix} 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix}}_N.$$

N nilpotent with $N^k = 0$. With " $a = \lambda I$ " and " $x = \lambda I + N$ "

$$f(J_k(\lambda)) = f(\lambda I) + \frac{f'(\lambda I)}{1!} N + \dots + \frac{f^{(k-1)}(\lambda I)}{(k-1)!} N^{k-1}$$

(Always a finite number of terms since $N^k = 0$.) Rule 2:

$$f(J_k(\lambda)) = f(\lambda)I + \frac{f'(\lambda)I}{1!} N + \dots + \frac{f^{(k-1)}(\lambda)I}{(k-1)!} N^{k-1}$$



Rule 4

Therefore rule 4 is:

$$f(J_k(\lambda)) = \begin{pmatrix} f(\lambda) & \frac{f'(\lambda)}{1!} & \frac{f''(\lambda)}{2!} & \cdots & \frac{f^{(k-1)}(\lambda)}{(k-1)!} \\ 0 & f(\lambda) & \frac{f'(\lambda)}{1!} & \cdots & \frac{f^{(k-2)}(\lambda)}{(k-2)!} \\ 0 & 0 & f(\lambda) & \cdots & \frac{f^{(k-3)}(\lambda)}{(k-3)!} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & f(\lambda) \end{pmatrix}$$

To compute $f(A)$ we need f and its derivatives (up to some order) on the eigenvalues of A . We say that f is defined on A if these can be computed.



Calculating $f(A)$

Two ways of computing matrix-polynomials

- 1 Directly compute monomials A^k using matrix multiplication.
- 2 Find Jordan form $J = S^{-1}AS$, compute $p(J_k(\lambda))$ for all blocks and $Sp(J)S^{-1}$.

For general functions the first alternative is infeasible due to large exponents in the Taylor expansion.

Theorem 9.1

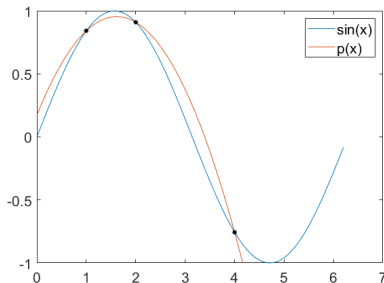
If $f(x), g(x)$ are defined A then

- 1 If $f(\lambda_i) = g(\lambda_i)$, $f'(\lambda_i) = g'(\lambda_i)$, ..., $f^{(s_i)}(\lambda_i) = g^{(s_i)}(\lambda_i)$ then $f(A) = g(A)$.
- 2 There exists a polynomial $p(x)$ of degree $< s$ such that $p(A) = f(A)$.
- 3 $(f + g)(A) = f(A) + g(A)$, $(fg)(A) = f(A)g(A)$.
- 4 $f(S^{-1}AS) = S^{-1}f(A)S$.

Calculating $f(A)$ using interpolation

Ex.

$$A = S \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix} S^{-1} \Rightarrow f(A) = S \begin{pmatrix} f(1) & 0 & 0 \\ 0 & f(2) & 0 \\ 0 & 0 & f(4) \end{pmatrix} S^{-1}$$



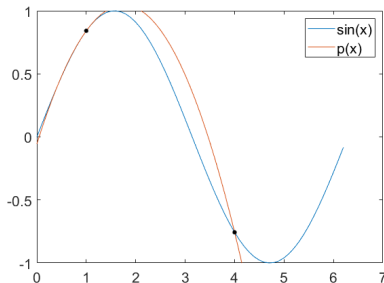
$$p(x) = -0.3003x^2 + 0.9687x + 0.1731$$
$$\sin(A) = -0.3003A^2 + 0.9687A + 0.1731I$$



Calculating $f(A)$ using interpolation

Ex.

$$A = S \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} S^{-1} \Rightarrow f(A) = S \begin{pmatrix} f(1) & f'(1) & 0 \\ 0 & f(1) & 0 \\ 0 & 0 & f(4) \end{pmatrix} S^{-1}$$



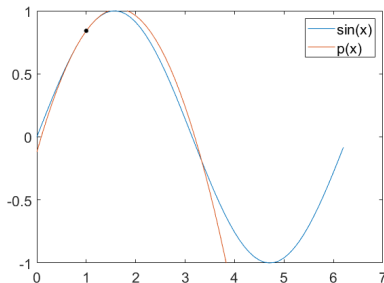
$$p(x) = -0.3577x^2 + 1.2557x - 0.0565$$
$$\sin(A) = -0.3577A^2 + 1.2557A - 0.0565I$$



Calculating $f(A)$ using interpolation

Ex.

$$A = S \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} S^{-1} \Rightarrow f(A) = S \begin{pmatrix} f(1) & f'(1) & f''(1)/2 \\ 0 & f(1) & f'(1) \\ 0 & 0 & f(1) \end{pmatrix} S^{-1}$$



$$p(x) = -0.4207x^2 + 1.3818x - 0.1196$$
$$\sin(A) = -0.4207A^2 + 1.3818A - 0.1196I$$

